Robust M3m3 H@te D3t3ct!on

Robustness of CLIP-fusion classifiers under naturalistic and adversarial perturbations

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Problem definition

- Hateful memes are compositional. Text or image alone may look benign, but together they can become hateful.
- Users obfuscate: leetspeak, spacing, compression, brightness shifts preserve meaning but can flip model predictions.
- Adversaries attack: FGSM/PGD perturbations expose whether the classifier is truly robust or only suitable for clean data.
- **⇒ Can a CLIP-based hate-meme detector stay reliable under both realistic and adversarial perturbations?**

Related works

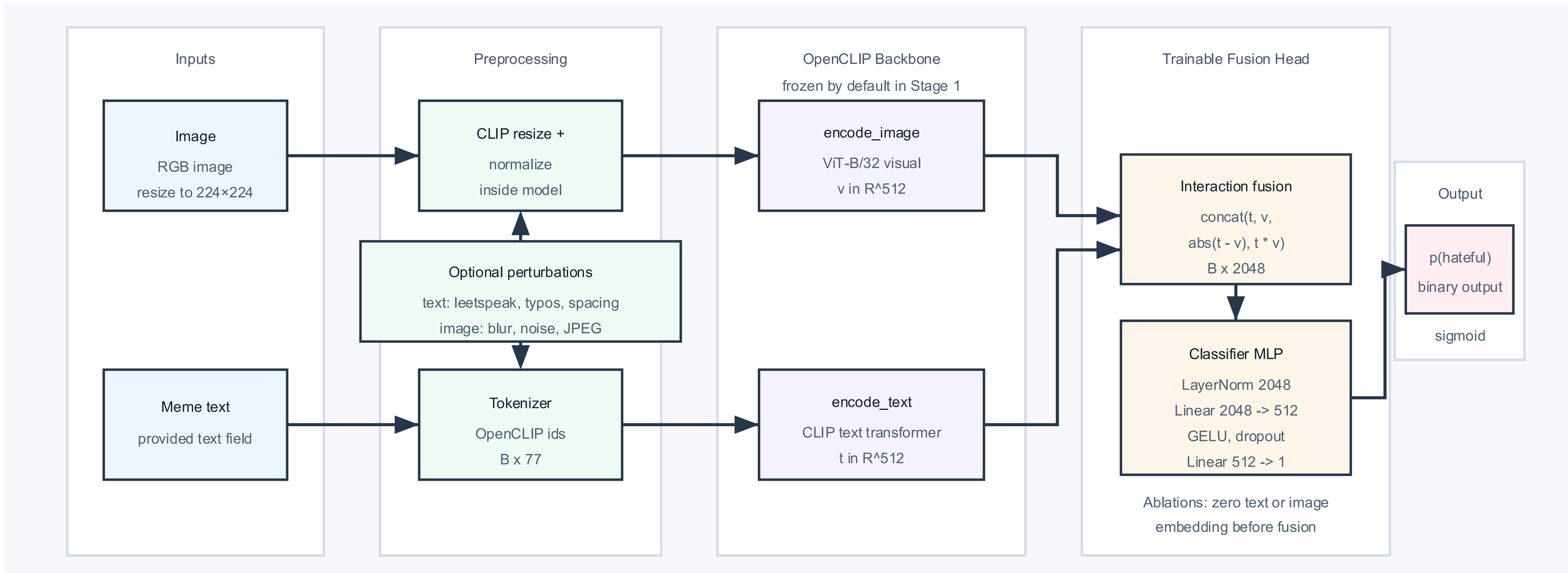
- Hateful Memes introduced controlled multimodal hate detection [1]
- CLIP-style dual encoders are practical, reproducible baselines for robustness studies [2]
- Fusion heads capture cross-modal interactions needed for compositional meme meaning [3]
- TRADES loss function [4]

Method

- We use frozen OpenCLIP ViT-B/32 image and text encoders with a lightweight fusion head

`concat(t, v, |t - v|, t ⊙ v) → LayerNorm → MLP → p(hateful)`

- Robust variants train on clean and perturbed views with BCE + consistency regularization



Perturbations
Common text perturbations (e.g. leetspeak, char deletion, spacing, etc.) and image perturbations (e.g. Gaussian noise/blur, compression, brightness shift, etc.) are applied at low, medium, and high severity levels

Datasets

Primary dataset:	Meta Hateful Memes
Supplementary:	MultiOFF, MAMI, HarMeme

Metrics

- Augonly: Uses only Cross-entropy against golden label in loss function
- KL: Adds Kullback-Leibler divergence measure in loss function
- KLdrop: Drops text encoding with probability p
- KLlowmed: KL that only trains on low and middle severity perturbations

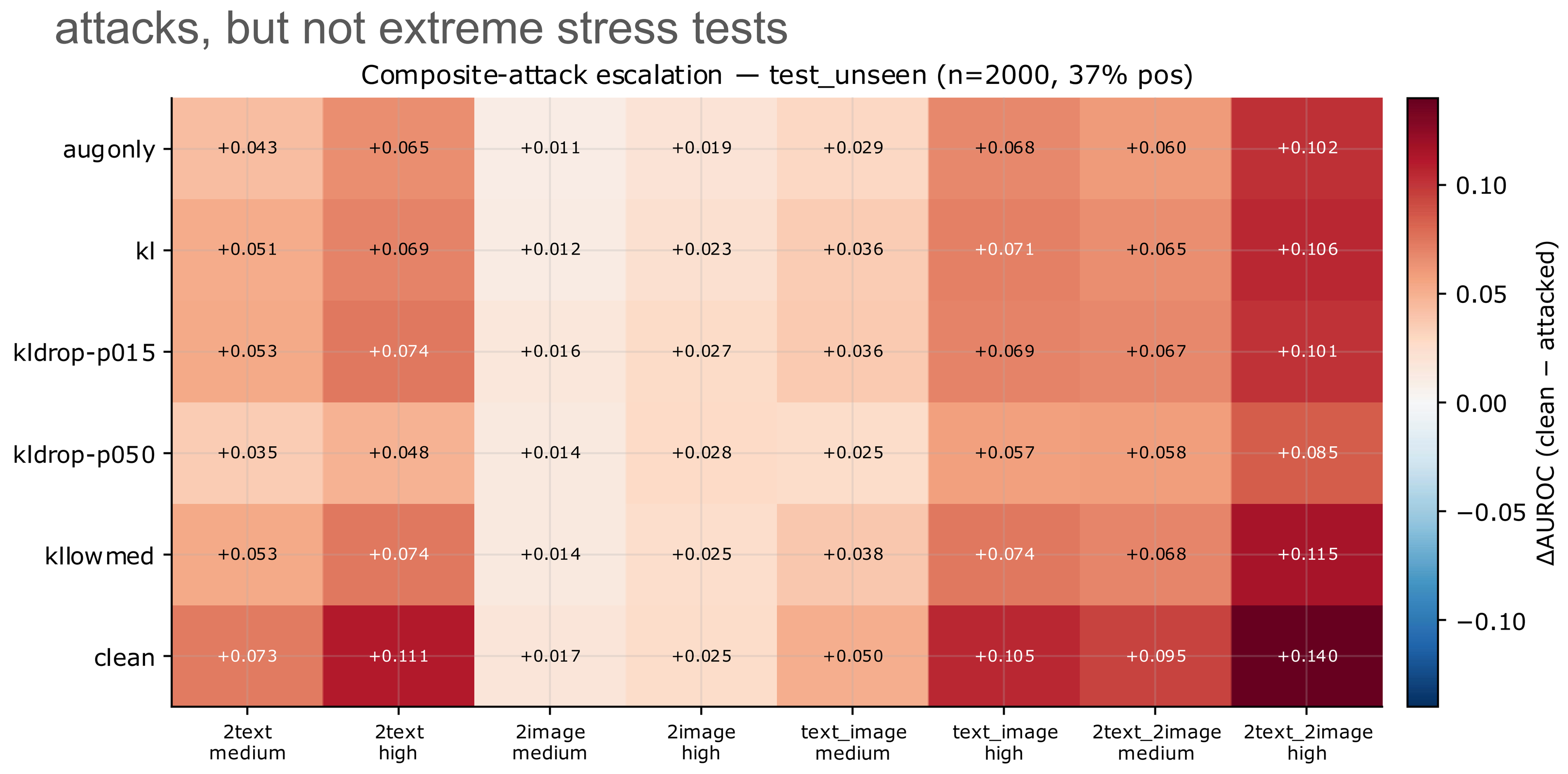
Loss Functions

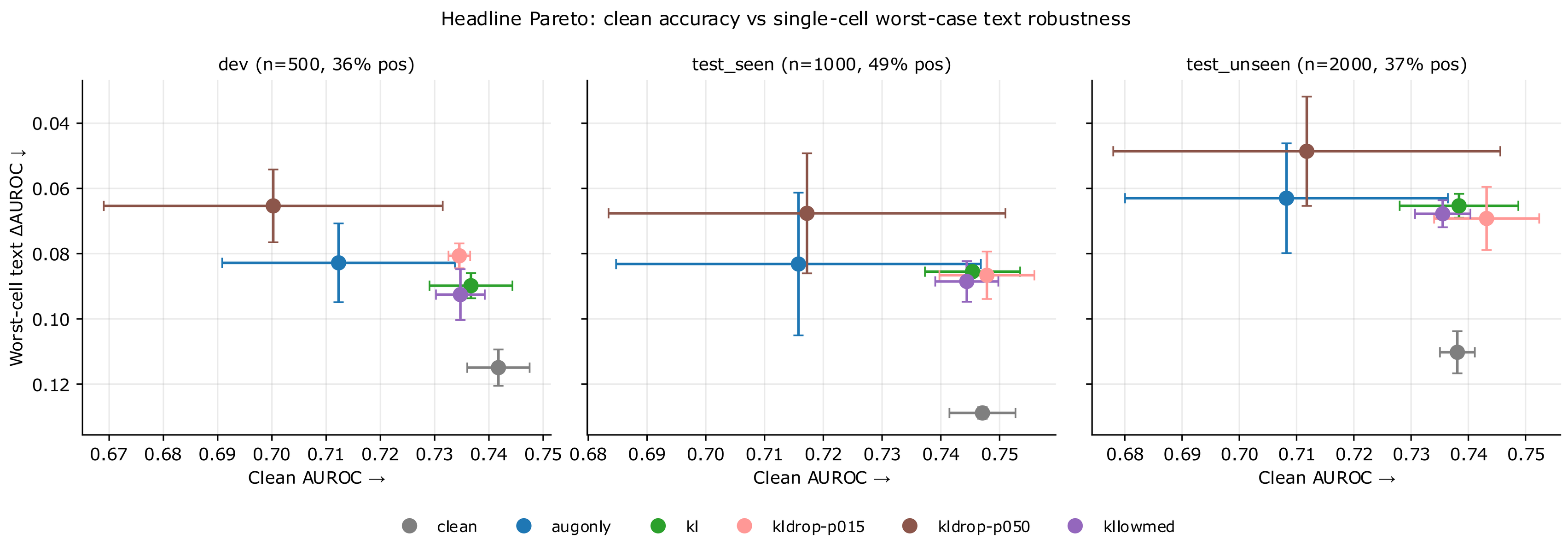
- Augonly:
  L = BCE(y, p_clean) + α · BCE(y, p_pert)
- KL & KLdrop:
  L = BCE(y, p_clean) + α · BCE(y, p_pert) + β · KL(p_clean ‖ p_pert)

Splits: Train: 8500 (HM train). Eval: dev (n=500, 36% pos), test_seen (n=1000, 49% pos) · test_unseen: (n=2000, 37% pos).
Metrics: AUROC (primary, no threshold) · macro-f1 · ASR · worst cell ΔAUROC across families.
Threshold: τ tuned on dev only (max macro-f1) · then frozen for test
Uncertainty: 3-seed mean ± σ reported; bootstrap 95% CI on per-class disagreement-count percentages.

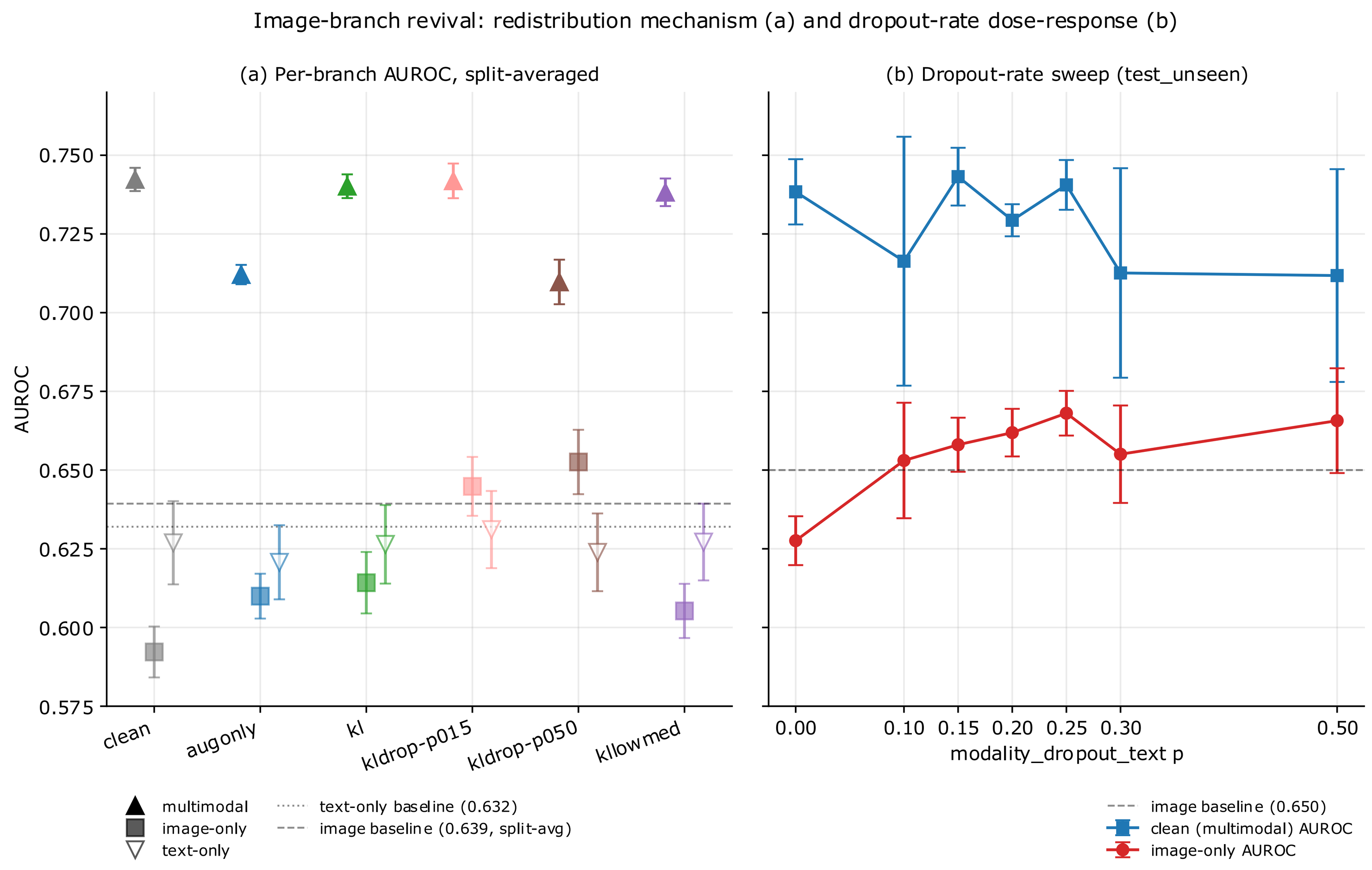
Validation / Results

- `kldrop-p015` and `kldrop-p025` are the best default region: clean AUROC is preserved while image-only AUROC improves
- `kldrop-p050` is the most robust under composite attacks, but pays a clean-AUROC cost
- Mixed severity composite attacks behave like realistic medium-severity attacks, but not extreme stress tests





Clean AUROC vs worst-case text robustness. `kl-drop-p015/p025` preserve clean performance while improving robustness



Text modality dropout revives the image branch: image-only AUROC rises while multimodal AUROC stays competitive

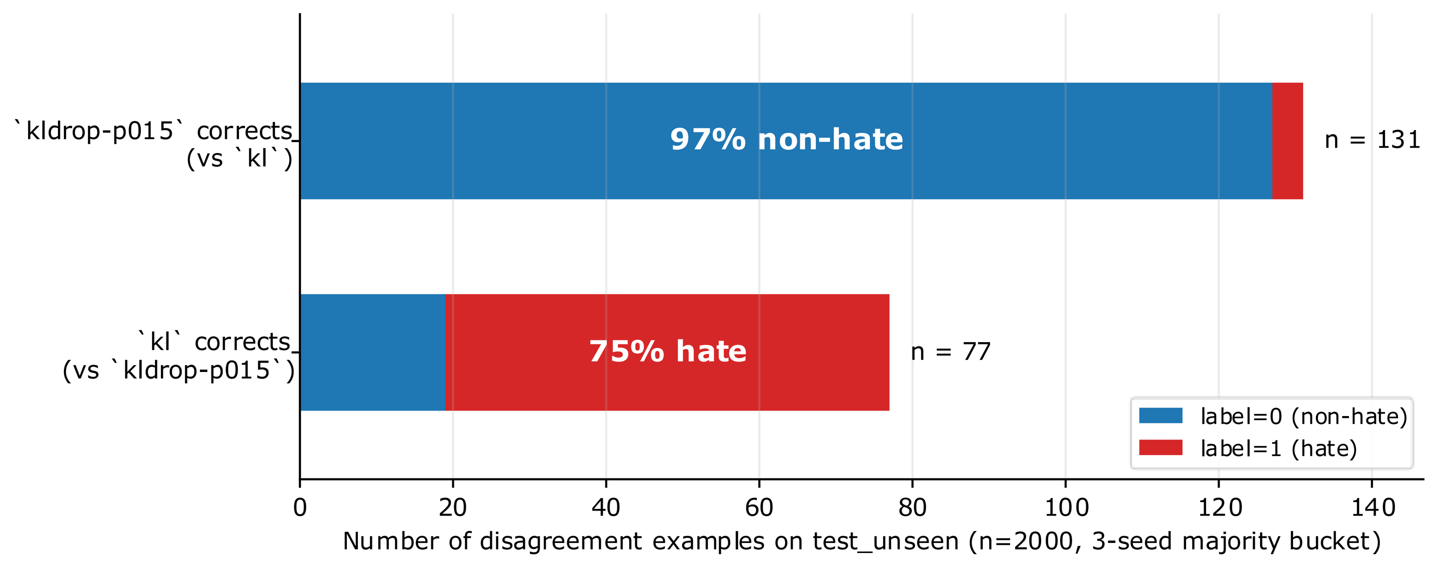


Image-branch revival cuts FPs on benign memes that mention a protected group. Hateful-class recall preserved within seed noise.

Adversarial Training
White-box FGSM/PGD exposes a separate failure mode. The clean model collapses under L∞ PGD, while L2 attacks are much less damaging.
Adversarial training improves L∞ robustness, but only when trained on a mix of clean and perturbed images; training only on perturbed images hurts clean performance more.
Hybrid and TRADES loss functions were used.

Model	Clean AUROC	PGD L∞ AUROC	PGD L2 AUROC
baseline	0.74	~0.04	0.7
adv-trained	0.73	0.5	0.73

Limitations

- Perturbations are label-preserving by design, but high-severity edits may become semantically ambiguous
- White-box PGD remains a hard threat; naturalistic robustness does not imply adversarial robustness
- The model is a research prototype, not a deployable moderation system

Conclusion / Future Improvements

Robust hate meme detection requires more than high clean accuracy. A CLIP-fusion model can be made substantially more reliable under realistic text/image obfuscations using perturbation training and modality dropout. The best default is **`kldrop-p015/p025`**, which restores image-branch usefulness without sacrificing clean AUROC. However, white-box adversarial attacks are a separate and harder robustness problem.
Future work includes grey and blackbox adversarial attacks, which are expected to have less negative impact but are more realistic. Furthermore, a combined model that is resistant to all attacks analyzed in this project to ensure overall robustness.

References

[1] D. Kiela et al., "The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 2611–2624.
[2] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in Proceedings of the 38th International Conference on Machine Learning, 2021, pp. 8748–8763.
[3] G. K. Kumar and K. Nandakumar, "Hate-CLIPper: Multimodal Hateful Meme Classification Based on Cross-modal Interaction of CLIP Features," in Proceedings of the Second Workshop on NLP for Positive Impact, 2022, pp. 171–183.